ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Environmental Chemistry

English Medium

EXERCISE # I

1. Which one is not the correct statement ?
Pollutions means :
(A) The presence of anything in the environment in excess of the required limit
(B) anything put into the environment which was not there in its natural state.
(C) direct or indirect changes in one or more components of the ecosystem which are not harmful to living organisms
(D) international release of any chemical substance in the environment with harmful effects

EN0001

2. Which is the main air pollutant ?
(A) CO_2 (B) CO (C) N_2 (D) S

EN0002

3. The acid rain possesses :
(A) Sulphuric acid (B) nitric acid (C) sulphurous acid (D) all of these

EN0003

4. Carbon monoxide (CO) is harmful to man because -
(A) It competes with O_2 for haemoglobin (B) It forms carbolic acid
(C) It generates excess CO_2 (D) It is carcinogenic

EN0004

5. Lung diseases are four times more in urban areas than rural areas. This is due to the presence of :-
(A) SO_2 (B) CO_2 (C) N_2 (D) Water-vapour

EN0005

6. Which one is not correct ?
Greenhouse effect :
(A) is due to high concentration of CO_2 in atmosphere
(B) is influenced by gases such as CH_4, O_3 and chlorofluorocarbons
(C) would result in the warming up of the earth
(D) would result in lowering the level of oceans due to high evaporation

EN0006

7. Which is not a green house gas -
(A) CFC's (B) Methane (C) H_2 (D) CO_2

EN0007

8. Green house effect is related to -
(A) Cultivation of green plants (C) Global wanning
(B) Cultivation of vegetables in houses (D) Global green algae

EN0008

9. Acid rains are produced by-
(A) Excess production of NH_3 by industry and coal gas
(B) Excess release of carbon monoxide by incomplete combustion
(C) Excess formation of CO_2 by combustion and animal respiration
(D) Excess NO_2 and SO_2, from burning of fossil fuels

EN0009

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10. If there was no CO_2 in the earth's atmosphere, the temperature of earth's surface would be -
(A) Less than the present
(B) The same
(C) Dependent on the amount of oxygen in the atmosphere
(D) Higher than the present

EN0010

11. The pH of acid rain water is :
(A) 1.2 (B) 3.1 (C) 5 (D) 6

EN0011

12. Which air pollutant is not released by scooters and cars ?
(A) SO_2 (B) Fly ash (C) Hydrocarbons (D) CO

EN0012

13. $CFCl_3$ is responsible for the decomposition of ozone to form oxygen. Which of the following reacts with ozone to form oxygen ?

(A) Cl_2 (B) Cl^- (C) F^- (D) $\dot{C}l$

EN0013

14. Pick up the correct statement
(A) CO which is major pollutant resulting from the combustion of fuels in automobiles plays a major role in photochemical smog
(B) Classical smog has an oxidizing character while the photochemical smog is reducing in character
(C) Photochemical smog occurs in day time whereas the classical smog occurs in early morning hours
(D) During formation of smog the level of ozone in the atmosphere goes down

EN0014

15. Air pollutants that produce photochemical smog -
(A) Ozone, chlorine and sulphur dioxide (B) Oxygen, chlorine and nitric acid
(C) Nitrous oxide, PAN and acrolein (D) CO_2, CO and SO_2

EN0015

16. P.A.N, stands for-
(A) Peroxy acetyl nitrite (B) Peroxy acetyl nitrate
(C) peroxy acetyl nitrile (D) Pyridine aceto-nitrite

EN0016

17. The photochemical smog can be suppressed by :
(A) nitrogen oxides (B) hydrocarbons (C) radical traps (D) formaldehyde

EN0017

18. The chief pollutants which are responsible for ozone depletion :-
(A) SO_2 (B) CO_2
(C) CO (D) Oxides of nitrogen and chlorofluorocarbons

EN0018

19. Taj mahal is threatened due to effect of :-
(A) Chlorine (B) Sulphur dioxide (C) Oxygen (D) Hydrogen

EN0019

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20. Ozone hole refers to :-
(A) Increased concentration of ozone
(B) Reduction in the thickness of ozone layer in troposphere
(C) Reduction in the thickness of ozone layer in stratosphere
(D) Hole in ozone layer

EN0020

21. One of the pollutants that is generally helping in the early degradation of the ozone layer is :-
(A) SO_2 (B) DDT (C) CO_2 (D) Freons

EN0021

22. In coming years, skin related disorders will become more common due to -
(A) Airpollution (B) Excessive use of detergents
(C) Depletion of ozone layer (D) Water pollution

EN0022

23. Often in water bodies subjected to sewage pollution, fishes die because of the-
(A) Foul smell
(B) Reduction in dissolved oxygen caused by microbial activity
(C) Clogging of their gills by solid substances
(D) Pathogens released by the sewage

EN0023

24. Which one is not a water pollutant ?
(A) Automobile exhaust (B) Plant nutrients
(C) Oxygen demanding wastes (D) Disease causing agents

EN0024

25. If BOD of river is high, it means that the river is :-
(A) Not polluted
(B) Very much polluted with inorganic chemicals
(C) Very much polluted with organic chemicals which are decomposed by micro-organisms
(D) Polluted with pesticides

EN0025

26. When huge amount of sewage is dumped in a river, the BOD will :-
(A) Increase (B) Remain unchanged
(C) Decrease (D) Increase or decrease

EN0026

27. By what method the quantity of organic pollutants in water can be determined
(A) By measuring BOD (B) By pH Measurement
(C) By transparency measurements (D) By measuring the change of colour/CFC

EN0027

28. A dental disease characterised by mottling of teeth is due to presence of a certain chemical element in drinking water. Which is the element -
(A) Fluorine (B) Mercury (C) Boron (D) Chlorine

EN0028

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29. Excess of nitrate in drinking water causes :-
(A) itai-itai syndrome
(B) Rickets syndrome
(C) Laxative effect
(D) Methemoglobinemia ('blue baby' syndrome)

EN0029

30. Eutrophication is caused by
(A) Increase nutrient concentration
(B) Petrochemical and fertilizer plant effluents
(C) NO_3^{-1} and SO_4^{-2} present in acid rains
(D) Mine effluents

EN0030

31. Continuous sewage flow into a stream would lead to-
(A) Increase in temperature
(B) Algal bloom
(C) Eutrophication
(D) Depletion of oxygen

EN0031

32. Indiscriminate use of DDT is undesirable because -
(A) It is harmful
(B) It is dcgradable
(C) It causes mutation
(D) It is accumulated in food chain

EN0032

33. Fluorosis, a bone disease, is caused by the presence of :-
(A) Pesticides in water
(B) Fluorides in water
(C) Carbon monoxide in air
(D) Sulphur dioxide in air

EN0033

34. Major source of methane in India is :
(A) fruit orchards
(B) sugar cane plantatton
(C) rice fields
(D) wheat fields

EN0034

35. Green chemistry means such reactions which :
(A) are related to the depletion of ozone layer
(B) produce colour during reactions
(C) study the reactions in plants
(D) reduce the use and production of hazardous chemicals

EN0035

36. Among the following, the one that is is not a green house gas is :-
(A) Nitrous oxide (B) Methane (C) Ozone (D) Sulphur dioxide

EN0036

37. Which oxide of nitrogen is not a common pollutant introduced into the atmosphere both due to natural and human activity?
(A) NO_2 (B) N_2O (C) N_2O_5 (D) NO

EN0037

38. Which of the following is a sink for CO?
(A) Haemoglobin
(B) Micro-organisms present in the soil
(C) Oceans
(D) Plants

EN0038

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EXERCISE # JEE-MAINS

1. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value
(1) slightly higher than that when the thunderstorm is not there **[AIEEE-2003]**
(2) uninfluenced by occurence of thunderstorm
(3) which depends on the amount of dust in air
(4) slightly lower than that of rain water without thunderstorm

EN0039

2. The smog is essentially caused by the presence of : **[AIEEE-2004]**
(1) O_2 and O_3 (2) O_3 and N_2
(3) Oxides of sulphur and nitrogen (4) O_2 and N_2

EN0040

3. Regular use of which of the following fertilizers increases the acidity of soil ? **[AIEEE-2007]**
(1) potassium nitrate (2) Urea
(3) Superphosphate of lime (4) Ammonium sulphate

EN0041

4. Identify the wrong statements in the following : **[AIEEE-2008]**
(1) Chlorofluorocarbons are responsible for ozone layer depletion
(2) Greenhouse effect is responsible for global warming
(3) Ozone layer does not permit infrared radiation from the sun to reach the earth
(4) Acid rains is mostly because of oxides of nitrogen and sulphur

EN0042

5. What is DDT among the following : **[JEE-MAINS-2012]**
(1) Greenhouse gas (2) A fertilizer
(3) Biodegradable pollutant (4) Non-biodegradable pollutant

EN0043

6. The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was :
(1) Methylisocyanate (2) Methylamine **[JEE-MAINS-2013]**
(3) Ammonia (4) Phosgene

EN0044

7. **Assertion :** Nitrogen and Oxygen are the main components in the atmosphere but these do not react to form oxides of nitrogen. **[JEE-MAINS-2015]**
Reason : The reaction between nitrogen and oxygen requires high temperature.
(1) Both assertion and reason are correct, and the reason is the correct explanation for the assertion
(2) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion
(3) The assertion is incorrect, but the reason is correct
(4) Both the assertion and reason are incorrect

EN0045

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8. The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000 ppb, 40 ppb, 100 ppm and 0.2 ppm, respectively. This water is unsuitable for drinking due to high concentration of : **[JEE-MAINS-2016]**

(1) Fluoride (2) Lead (3) Nitrate (4) Iron

EN0046

9. A water sample has ppm level concentration of following anions $F^- = 10$; $SO_4^{2-} = 100$; $NO_3^- = 50$ the anion/anions that make / makes the water sample unsuitable for drinking is / are :

(1) only NO_3^- (2) both SO_4^{2-} and NO_3^- **[JEE-MAINS-2017]**

(3) only F^- (4) only SO_4^{2-}

EN0047

10. Identify the pollutant gases largely responsible for the discoloured and lustreless nature of marble of the Taj Mahal **[JEE-MAINS-2017]**

(1) O_3 and CO_2 (2) CO_2 and NO_2 (3) SO_2 and O_3 (4) SO_2 and NO_2

EN0048

11. Which of the following is a set of green house gases ? **[JEE-MAINS-2017]**

(1) O_3, N_2, CO_2, NO_2 (2) CH_4, O_3,N_2, SO_2

(3) CO_2, CH_4, N_2O, O_3 (4) O_3, NO_2, SO_2, Cl_2

EN0049

12. Biochemical Oxygen Demand (BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is correct - **[JEE-MAINS ONLINE-2018]**

(1) Anaerobic bacteria increase the BOD value

(2) Aerobic bacteria decrease the BOD value

(3) Polluted water has BOD value higher than 10 ppm

(4) Clean water has BOD value higher than 10 ppm

EN0050

13. The recommended concentration of fluoride ion in drinking water is up to 1 ppm as fluoride ion is required to make teeth enamel harder by converting $[3Ca_3(PO_4)_2 \cdot Ca(OH)_2]$ to : **[JEE-MAINS-2018]**

(1) $[3(CaF_2) \cdot Ca(OH)_2]$

(2) $[3(Ca_3(PO_4)_2 \cdot CaF_2]$

(3) $[3(Ca(OH)_2] \cdot CaF_2]$

(4) $[CaF_2]$

EN0051

14. A water sample has ppm level concentration of the following metals: Fe= 0.2; Mn = 5.0; Cu = 3.0; Zn = 5.0. The metal that makes the water sample unsuitable drinking is : **[JEE-MAINS-2019]**

(1) Zn (2) Fe (3) Mn (4) Cu

EN0052

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15. Which of the following conditions in drinking water causes methemoglobinemia ?

(1) > 50ppm of load (2) > 100 ppm of sulphate **[JEE-MAINS-2019]**

(3) > 50 ppm of chloride (4) > 50 ppm of nitrate

EN0053

16. The reaction that is NOT involved in the ozone layer depletion mechanism is the stratosphere is:

(1) $HOCl(g) \xrightarrow{h\upsilon} \dot{O}H(g) + \dot{C}l(g)$ **[JEE-MAINS-2019]**

(2) $CF_2Cl_2(g) \xrightarrow{uv} \dot{C}l(g) + \dot{C}F_2Cl(g)$

(3) $CH_4 + 2O_3 \rightarrow 3CH_2 = O + 3H_2OP$

(4) $Cl\dot{O}(g) + O(g) \rightarrow \dot{C}l(g) + O_2(g)$

EN0054

17. The compound that is NOT a common component of photochemical smog is :**[JEE-MAINS-2019]**

(1) O_3 (2) CH_2=CHCHO

(3) CF_2Cl_2 (4) $H_3C-\overset{\overset{O}{\|}}{C}-OONO_2$

EN0055

18. The upper stratosphere consisting of the ozone layer protects us from the sun's radiation that falls in the wavelength region of: **[JEE-MAINS-2019]**

(1) 600-750 nm (2) 0.8-1.5 nm

(3) 400-550 nm (4) 200-315 nm

EN0056

19. Biochemical Oxygen Demand (BOD) is the amount of oxygen required (in ppm):

(1) by anaerobic bacteria to breakdown inorganic waste present in a water body.

(2) for the photochemical breakdown of waste present in 1 m^3 volume of a water body.

(3) by bacteria to break-down organic waste in a certain volume of a water sample.

(4) for sustaining life in a water body. **[JEE-MAINS-2020]**

EN0057

20. Among the gases (a) - (e), the gases that cause greenhouse effect are : **[JEE-MAINS-2020]**

(a) CO_2 (b) H_2O (c) CFCs (d) O_2 (e) O_3

(1) (a), (b), (c) and (d)

(2) (a), (c), (d) and (e)

(3) (a) and (d)

(4) (a), (b), (c) and (e)

EN0058

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21. Given below are two statements : **[J-Main 2021]**

Statement I: The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life. **EN0059**

Statement II: The optimum value of BOD is 6.5 ppm.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

22. Given below are two statements: **[J-Main 2021]**

Statement I : An allotrope of oxygen is an important intermediate in the formation of reducing smog.

Statement II : Gases such as oxides of nitrogen and sulphur present in troposphere contribute to the formation of photochemical smog. **EN0060**

In the light of the above statements, choose the correct answer from the options given below:

(1) Both statement I and Statement II are false

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Statement I is false but Statement II is true

23. Reducing smog is a mixture of: **[J-Main 2021]**

(1) Smoke, fog and O_3

(2) Smoke, fog and SO_2

(3) Smoke, fog and CH_2=CH–CHO

(4) Smoke, fog and N_2O_3 **EN0061**

24. The satements that are TRUE : **[J-Main 2021]**

(A) Methane leads to both global warming and photochemical smog

(B) Methane is generated from paddy fields

(C) Methane is a stronger global warming gas than CO_2

(D) Methane is a part of reducing smog.

Choose the most appropriate answer from the options given below :

(1) (A), (B), (C) only (2) (A) and (B) only

(3) (B), (C), (D) only (4) (A), (B), (D) only

EN0062

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25. Given below are two statements : **[J-Main 2021]**

Statement I : Chlorofluoro carbons breakdown by radiation in the visible energy region and release chlorine gas in the atmosphere which then reacts with stratospheric ozone. **EN0063**

Statement II : Atmospheric ozone reacts with nitric oxide to give nitrogen and oxygen gases, which add to the atmosphere.

For the above statements choose the correct answer from the options given below :

(1) **Statement I** is incorrect but **statement II** is true

(2) Both **statement I** and **II** are false

(3) **Statement I** is correct but **statement II** is false

(4) Both **statement I** and **II** are correct

26. Correct statement about photo-chemical smog is : **[J-Main 2022]**

(1) It occurs in humid climate. **EN0064**

(2) It is a mixture of smoke, fog and SO_2

(3) It is reducing smog.

(4) It results from reaction of unsaturated hydrocarbons.

27. The measured BOD values for four different water samples (A-D) are as follows: **[J-Main 2022]**

A = 3 ppm: B=18 ppm: C-21 ppm: D=4 ppm. The water samples which can be called as highly polluted with organic wastes, are

(1) A and B (2) A and D

(3) B and C (4) B and D

EN0065

28. The eutrophication of water body results in : **[J-Main 2022]**

(1) loss of Biodiversity

(2) breakdown of organic matter

(3) increase in biodiversity

(4) decrease in BOD.

EN0066

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ANSWER KEY

EXERCISE # I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	A	A	D	C	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	D	C	C	B	C	D	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	B	A	C	A	A	A	D	A
Que.	31	32	33	34	35	36	37	38		
Ans.	D	D	B	C	D	D	C	B		

EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	4	3	4	1	1	3	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	2	3	4	3	3	4	3	4
Que.	21	22	23	24	25	26	27	28		
Ans.	3	1	2	1	2	4	3	1		

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